

Multiplication Learning Game

1. Introduction

The Multiplication Learning Game is a console-based educational application designed to help users practice multiplication skills. Users are challenged with multiplication questions, and their performance is tracked across different difficulty levels. The game adjusts its difficulty based on the user's performance and provides feedback on their progress.

2. System Overview

The game provides the following functionalities:

- Start Game: Begin a new game session where users answer multiplication questions.
- Help: Access instructions and rules for playing the game.
- Show High Score: View the highest score achieved in the game.
- Exit: Close the game application.

3. Features and Functionality

3.1. Start Game

Purpose: To initiate a new game session where the user answers multiplication questions.

Process:

1. Name Entry: The user is prompted to enter their name.
2. Question Generation: The game generates multiplication questions based on the current difficulty level.
3. Answer Checking: User responses are checked for correctness.
4. Scoring: Correct and incorrect answers are tracked.
5. Difficulty Adjustment: After 3 consecutive correct answers, the difficulty level increases. If the user answers 3 questions wrong in a row, the game ends.
6. Feedback: The user receives feedback on their performance and the game concludes after 20 questions or upon "Game Over."

3.2. Help

Purpose: To provide instructions and guidance on how to play the game. Process:

1. Game Description: The game explains that it is a multiplication learning tool.
2. Gameplay Instructions: Users are informed about answering multiplication questions and the rules for advancing difficulty levels.

3.3. Show High Score

Purpose: To display the highest score achieved in the game. Process:

1. High Score Display: The game shows the current high score based on the best performance recorded.

3.4. Exit

Purpose: To close the game application. Process:

1. Exit Confirmation: The game confirms that the user wants to exit and terminates the application.

4. Error Handling

- Invalid Menu Option: If the user selects an invalid option, they are prompted to choose a valid option again.
- Game Over: The game ends if the user gets 3 consecutive wrong answers, and a "Game Over" message is displayed.

5. Code Details

- Random Number Generation: The game uses the `<cstdlib>` library for `std::rand()` and `std::srand()` to generate random numbers for question creation.
- Time Seeding: The `<ctime>` library provides `std::time()` for seeding the random number generator, ensuring that each game session generates unique questions.

Workflow

- Initialization
- Seed the random number generator.
- Display the home screen with menu options.
- Menu Selection
- User selects an option from the menu.
- Start Game
- Prompt for the player's name.
- Initialize game variables including successive_correct and successive_wrong.
- For each of the 20 questions:
 - Generate a multiplication question based on the current difficulty level.
 - Check the user's answer:
 - If correct:
 - Increment successive_correct.
 - Reset successive_wrong.
 - Update the score.
 - Check if the user has answered 3 questions correctly in a row at the 3rd difficulty level. If so, display a winning message and end the game.
 - If incorrect:
 - Increment successive_wrong.
 - Reset successive_correct.
 - Update the score.
 - If the user answers incorrectly 3 times in a row, end the game.
 - Check for a difficulty level change:
 - If the user answers 3 questions correctly in a row, increase the difficulty level and reset successive_correct.
- Game Over
- Display a message if the user has answered 3 questions incorrectly in a row.
- Calculate the final score.
- Provide feedback based on the final score.
- Show the high score.
- Help
- Display game instructions.
- Show High Score
- Display the highest score recorded.
- Exit
- End the program and display a farewell message.

Sample Output

```
Welcome to the Multiplication Learning Game!
1. Start Game
2. Help
3. Show High Score
4. Exit
Please choose an option (1-4): 1
Enter your name: Hanin
How much is 1 times 3? 1
Wrong answer.
How much is 5 times 8? 1
Wrong answer.
How much is 1 times 7? 1
Wrong answer.
Game Over. You've made 3 successive wrong answers.
Please ask your teacher for extra help.
High Score: 0

Process returned 0 (0x0)   execution time : 6.259 s
Press any key to continue.
```

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